

MID TERM EXAMINATION

Subject: d, a, t, a, s, t, r, u, c, t, u, r, e, s

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 40 | Date: 12-03-2026

Total Questions: 11 | Answer all questions.

Section 1: MCQ (5 Questions - 10 Marks)

Q1. [2 marks] (Medium)

Mid Term Examination | Course: Data Structures (CS101)

Q2. [2 marks] (Easy)

Which of the following is a linear data structure?

Options:

A) Tree B) Graph C) Array D) Hash Table

Q3. [2 marks] (Easy)

In C/C++, what operator is used to dereference a pointer, i.e., access the value stored at the memory address it points to?

Options:

A) & B) * C) -> D) .

Q5. [2 marks] (Medium)

What is the time complexity to insert a new node at the beginning of a singly linked list?

Options:

A) $O(N)$ B) $O(\log N)$ C) $O(1)$ D) $O(N \log N)$

Q9. [2 marks] (Hard)

Which of the following statements is true for a Binary Search Tree (BST)?

Options:

A) The left child of a node is always greater than the node itself. B) The right child of a node is always smaller than the node itself. C) All leaf nodes are always at the same depth. D) For any node, all values in its left subtree are smaller than the node's value, and all values in its right subtree are greater.

Section 2: Short Answer (6 Questions - 30 Marks)

Q4. [5 marks] (Easy)

Explain the concept of "contiguous memory allocation" as it applies to arrays.

Q6. [5 marks] (Medium)

Differentiate between a Stack and a Queue data structure based on their primary access mechanisms.

Q7. [5 marks] (Medium)

State two distinct advantages of using a Doubly Linked List over a Singly Linked List.

Q8. [5 marks] (Medium)

List two common real-world applications where the Stack data structure is prominently used.

Q10. [5 marks] (Hard)

Describe the process of an Inorder traversal for a Binary Tree. Provide a small example of a tree and its Inorder traversal output.

Q11. [5 marks] (Hard)

Explain why recursion can sometimes be less efficient or even problematic compared to an iterative approach, especially in scenarios involving deep traversals like a very long linked list.

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